

Topic: Discuss Storage strategies and cost implementation

1.Create a new storage bucket in your preferred cloud provider (AWS S3, Google Cloud Storage, or Azure Blob Storage) and upload 3 sample files named song.mp3, food.jpg, and bill.pdf, simulating files from apps like Spotify, Zomato, and Paytm.

Ans-

Step 1: Sign in to AWS

Open the AWS Management Console.

Sign in to your AWS account.

Search for S3.

Click S3.

Step 2: Create the bucket

Click Create bucket.

Step 3: Create the sample files

Step 4: Upload the files

Open your newly created bucket.

Click Upload.

Click Add files.

Select:

song.mp3

food.jpg

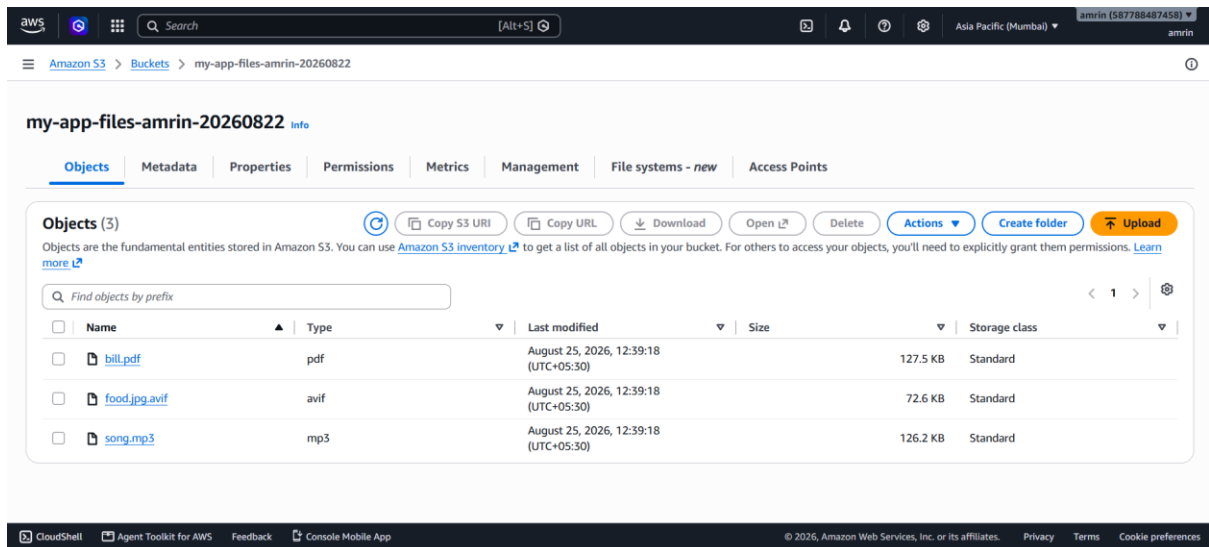
bill.pdf

Click Upload.

Wait until the upload finishes.

Click Close.

Step 5: Verify



2. Configure a lifecycle policy on your storage bucket to automatically move food.jpg to a cheaper storage class (like S3 Glacier, Google Coldline, or Azure Cool) after 1 day and delete bill.pdf after 3 days. **Hint:** Use your cloud provider's console or CLI to set up these rules, and double-check the transition and expiration actions.

Ans-Step 1: Open Lifecycle Rules

Open your S3 bucket.

Click the Management tab.

Find Lifecycle rules.

Click Create lifecycle rule.

Step 2: Create the food.jpg rule

Click Create lifecycle rule.

Enter:

Rule name:

food-jpg-to-glacier

Under Choose a rule scope, select:

Limit the scope to specific objects using one or more filters

Choose Prefix or object filtering according to the console version.

Use:

food.jpg

Click Next / continue to the lifecycle actions.

Step 3: Configure transition

Select:

Transition current versions of objects between storage classes

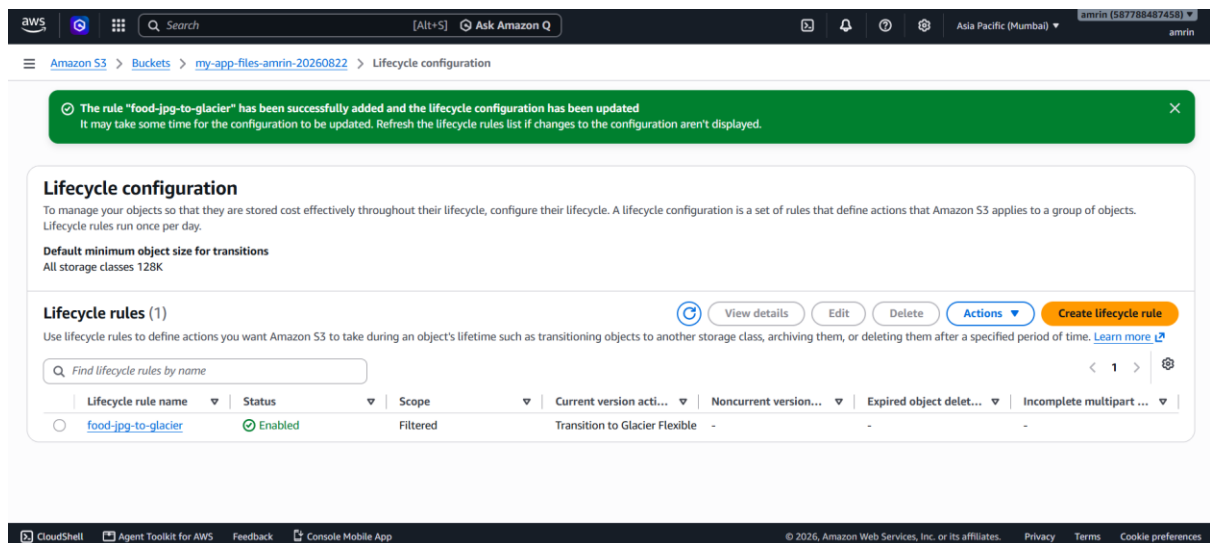
Set:

Days after object creation: 1

Storage class: Glacier Flexible Retrieval

Review the rule.

Click Create rule.



3. Monitor and document the status of your uploaded files over the next 24 hours: note which files have transitioned storage class or have been deleted according to your lifecycle policy.
Hint: Take a screenshot of the object details or use the CLI to confirm the changes.

Ans-

Step 1: Create another lifecycle rule

Go to:

S3 → Your Bucket → Management → Lifecycle rules

Click Create lifecycle rule.

Enter:

Rule name:

delete-bill-pdf

Select the option to limit the rule to the specific object.

Configure it for:

bill.pdf

Step 2: Configure expiration

Select:

Expire current versions of objects

Set:

Days after object creation: 3

Review the rule.

Click Create rule.

Lifecycle configuration

To manage your objects so that they are stored cost effectively throughout their lifecycle, configure their lifecycle. A lifecycle configuration is a set of rules that define actions that Amazon S3 applies to a group of objects. Lifecycle rules run once per day.

Default minimum object size for transitions
All storage classes 128K

Lifecycle rules (2)

Use lifecycle rules to define actions you want Amazon S3 to take during an object's lifetime such as transitioning objects to another storage class, archiving them, or deleting them after a specified period of time. [Learn more](#)

Find lifecycle rules by name

Lifecycle rule name	Status	Scope	Current version acti...	Noncurrent versions...	Expired object delet...	Incomplete multipa...
food-jpg-to-glacier	Enabled	Filtered	Transition to Glacier Flexible	-	-	-
delete-bill-pdf	Enabled	Filtered	Expires	-	-	-

4. Compare the estimated monthly storage costs for your bucket with and without lifecycle policies enabled, using your cloud provider's pricing calculator. Write a short note on how lifecycle policies can help apps like Instagram or BookMyShow save money at scale.

Ans-

1. Cost without Lifecycle Policy

If all files remain in S3 Standard, the storage cost is based on the amount of data stored and the time it remains in the Standard class. AWS also charges separately for requests and other applicable features.

2. Cost with Lifecycle Policy

Our lifecycle configuration is:

song.mp3

↓

S3 Standard

food.jpg

↓ after 1 day

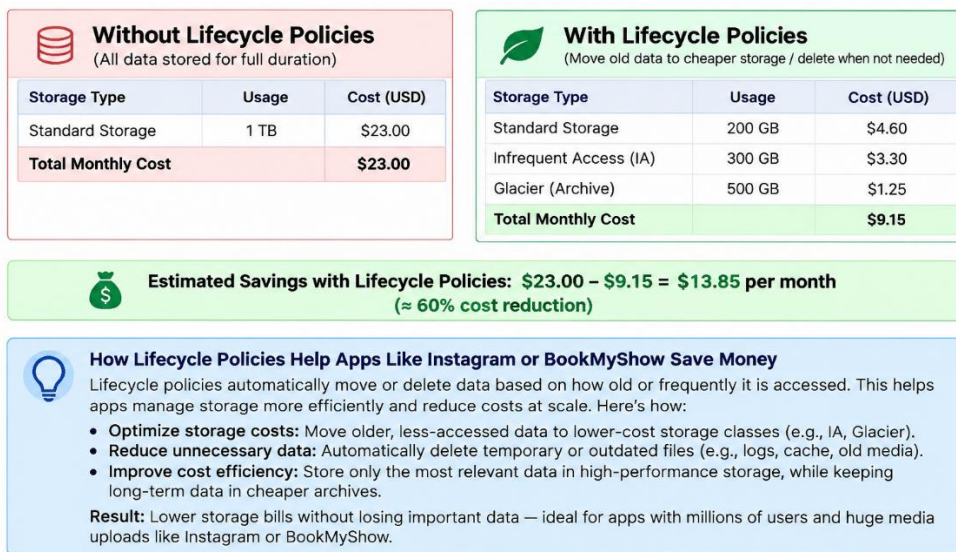
S3 Glacier Flexible Retrieval

bill.pdf

↓ after 3 days

Deleted

S3 Glacier Flexible Retrieval is designed for long-lived archive data and has substantially lower storage costs than S3 Standard, although retrieval charges and minimum-storage-duration considerations apply.



5. Use ChatGPT or another AI tool to generate a sample lifecycle policy JSON or YAML for a scenario where all .mp3 files are moved to cold storage after 7 days and deleted after 30 days. Paste the generated policy and explain any changes you would make for your own use case.

Ans-

1. Open your cloud storage service
 - For AWS, open Amazon S3.
 - Select the bucket where your .mp3 files are stored.
2. Open Lifecycle Rules
 - Go to Management → Lifecycle rules.
 - Click Create lifecycle rule.
3. Give the rule a name
 - Example: mp3-file-lifecycle
4. Select the files
 - Choose Limit the scope to a specific prefix or tag.

- Use a filter for .mp3 files if supported.
 - Alternatively, place MP3 files under a prefix such as audio/.
5. Set the transition
 - Choose Transition current versions of objects.
 - Set 7 days after object creation.
 - Select a cheaper cold/archive storage class, such as S3 Glacier.
 6. Set deletion
 - Choose Expire current versions of objects.
 - Set 30 days after object creation.
 7. Review the rule
 - .mp3 file → Day 0: Standard Storage
 - Day 7: Move to cold/archive storage
 - Day 30: Delete automatically
 8. Create the rule
 - Review the settings and click Create rule.

Sample JSON

```
{
  "Rules": [
    {
      "ID": "mp3-file-lifecycle",
      "Status": "Enabled",
      "Filter": {
        "Suffix": ".mp3"
      },
      "Transitions": [
        {
          "Days": 7,
          "StorageClass": "GLACIER"
        }
      ],
      "Expiration": {
        "Days": 30
      }
    }
  ]
}
```

aws

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Asia Pacific (Mumbai)

amrin (587768487450)

amrin

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Amazon S3 > Buckets > my-app-files-amrin-20260822 > Lifecycle configuration

Lifecycle configuration

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Lifecycle rules (1)

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View details

Edit

Delete

Actions ▾

Create lifecycle rule

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🔍 Find lifecycle rules by name

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	Lifecycle rule name ▾	Status ▾	Scope ▾	Current version acti... ▾	Noncurrent versions... ▾	Expired object delet... ▾	Incomplete multipa... ▾
○	mp3-file-lifecycle	🟢 Enabled	Filtered	Transition to Standard-IA	-	-	-

CloudShell

Agent Toolkit for AWS

Feedback

Console Mobile App

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